

## 8.5 RS232/V24 Operation

### 8.5.1 Hardware Configuration

The interface comes with a 9-pin male D-connector, that via a DB-9 female to DB-25 male adapter, complies with the RS232(EIA)- and the V24 (CCITT) standard.

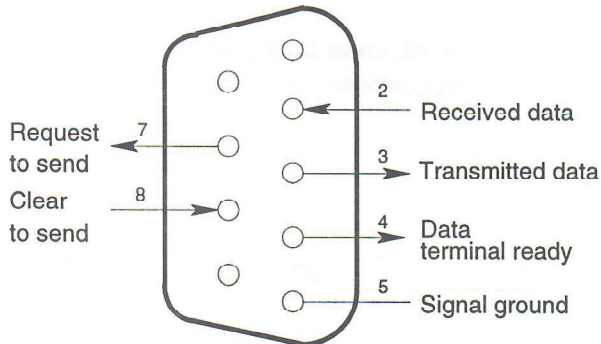


Fig. 8-3 Connector pin-out, RS232/V24

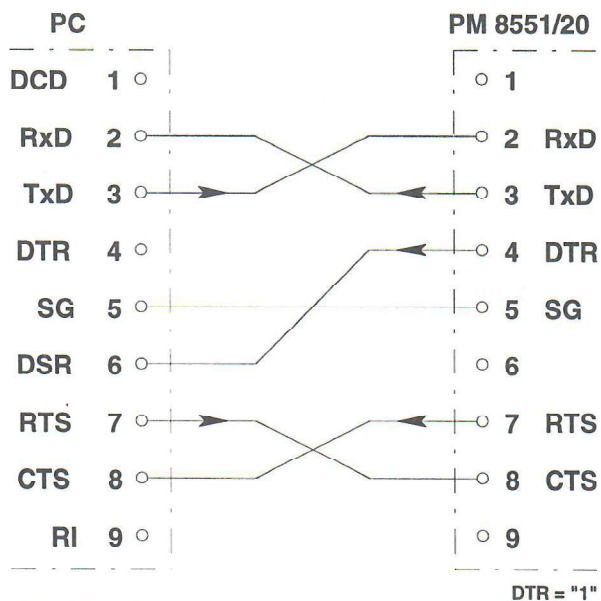


Fig. 8-4 Configuration of cable between PC and PM 8551/20.

The hardware configurationals include:

Communication settings:

- Baud rate
- Character length
- Parity selection

### 1. Communication settings:

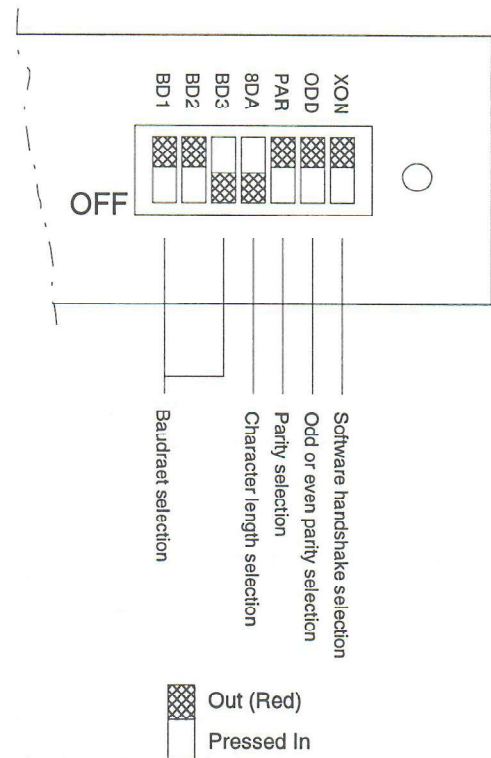


Fig. 8-5 Communication setting, SW1

| BD1        | BD2        | BD3       | Baudrate    |
|------------|------------|-----------|-------------|
| off        | off        | off       | 110         |
| on         | off        | off       | 150         |
| off        | on         | off       | 300         |
| on         | on         | off       | 600         |
| <b>off</b> | <b>off</b> | <b>on</b> | <b>1200</b> |
| on         | off        | on        | 2400        |
| off        | on         | on        | 4800        |
| on         | on         | on        | 9600        |

Table 8-6 Baudrate setting

| SWITCH     | FUNCTION                                                                       |
|------------|--------------------------------------------------------------------------------|
| <b>8DA</b> | <b>on</b> : 8 data bits character length<br>off : 7 data bits character length |
| <b>PAR</b> | on : Parity on<br><b>off</b> : <b>Parity off</b>                               |
| <b>ODD</b> | on : Odd parity<br><b>off</b> : <b>Even parity</b>                             |
| <b>XON</b> | <b>Must be in off position</b>                                                 |

Table 8-7 Character length, Parity and Handshake setting.

**Bold** = Standard Factory Setting

## 8.5.2 Interface Programming

The RS232/V24 remote interface can via the controller be configured with regard to:

–Record separator

Furthermore, it can run a number of 'Escape functions'.

The programming of the interface must follow the syntax illustrated on page 8-1.

### 1. Separator programming

The interface can be programmed to accept single and double separators. At power on the single separator "LF" is default setting (ISO code 10). When the interface sends data to the controller, the data stream is concluded with the programmed separator(s).

#### Definition of record separator(s).

| HEADER | DATA                                                                                                                                                                                                                                                             | SUFFIX | RETURNS |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| SPR    | n1 (,n2)<br>Separator 1 : n1<br>Separator 2 : n2<br>n1 and n2 are the decimal equivalent (ISO code) of the desired separator(s).<br>ESC (ISO 27), ETX (ISO 03) and EOT (ISO 04) are not allowed as separator.<br>Default separator at power on is "LF" (ISO 10). | none   |         |

Table 8-8 Programming, separator(s)

#### Example:

- "SPR 13" defines "CR" as input/output separator.
- "SPR 13, 10" defines the sequence "CR LF" as input/output separator.

### 2. Escape functions

| ESCAPE (n) | Function              |
|------------|-----------------------|
| 1          | Goto local            |
| 2          | Goto remote           |
| 3          | Goto local and unlock |
| 4          | Device clear          |
| 5          | Local lock out        |
| 7          | Status request        |
| 8          | Device trigger        |

Table 8-9 Escape functions

When using the "ESCAPE 7" function, a status byte will be send to the controller:

| Bit | Condition |
|-----|-----------|
| 7   | EXT       |
| 6   | RQS       |
| 5   | AB        |
| 4   | BSY       |
| 3   | EF3       |
| 2   | EF2       |
| 1   | EF1       |
| 0   | EF0       |

Table 8-10 Status byte

EXT:

not used

RQS:

not used

AB:

Abnormal condition

|   | AB                                                 |
|---|----------------------------------------------------|
| 0 | Normal (no error)                                  |
| 1 | Abnormal (error condition as specified by EF0-EF3) |

BSY:

Busy

EF0-EF3:

Message flags

| AB | EF3-EF0 | Dec. | Message                  |
|----|---------|------|--------------------------|
| 0  | 0001    | 1    | Message string available |
| 1  | 1100    | 12   | Number error             |
| 1  | 1101    | 13   | Execute error            |
| 1  | 1110    | 14   | Head/suffix error        |
| 1  | 1111    | 15   | Syntax error             |

### 8.5.3 Device programming, RS232/V24

A list of the available programming commands and a description of the programming syntax to be used can be found in paragraph 8.4